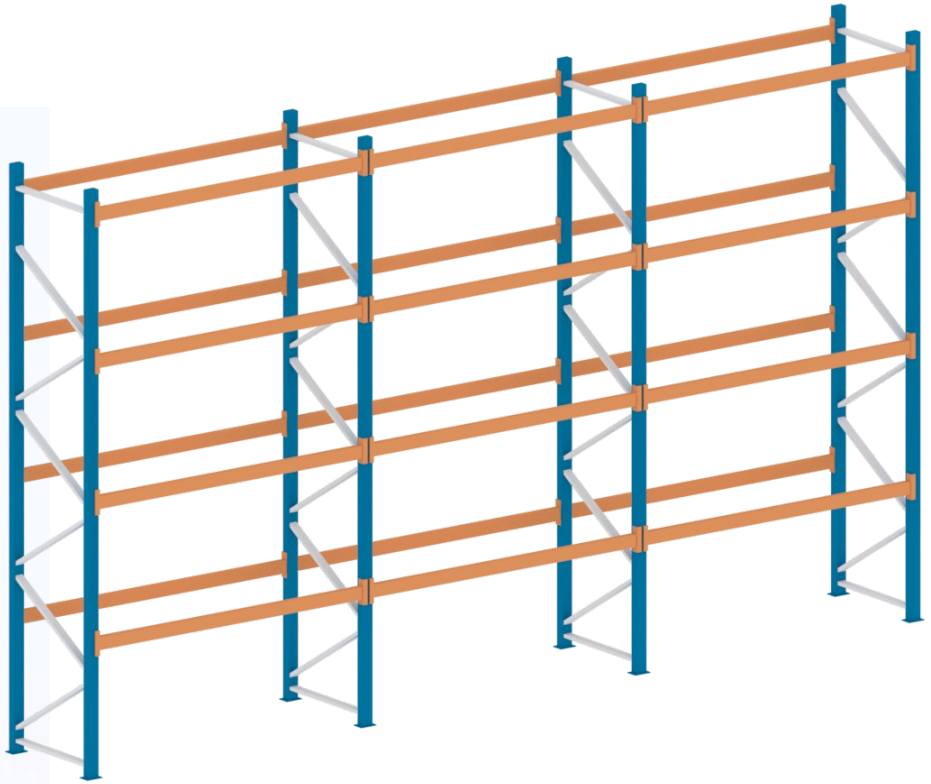
ANATOMY OF A RACK

Upright:

Uprights + Diagonal =
Frame:

Frames + Beam =
Bay:

Many Bays =
Row:



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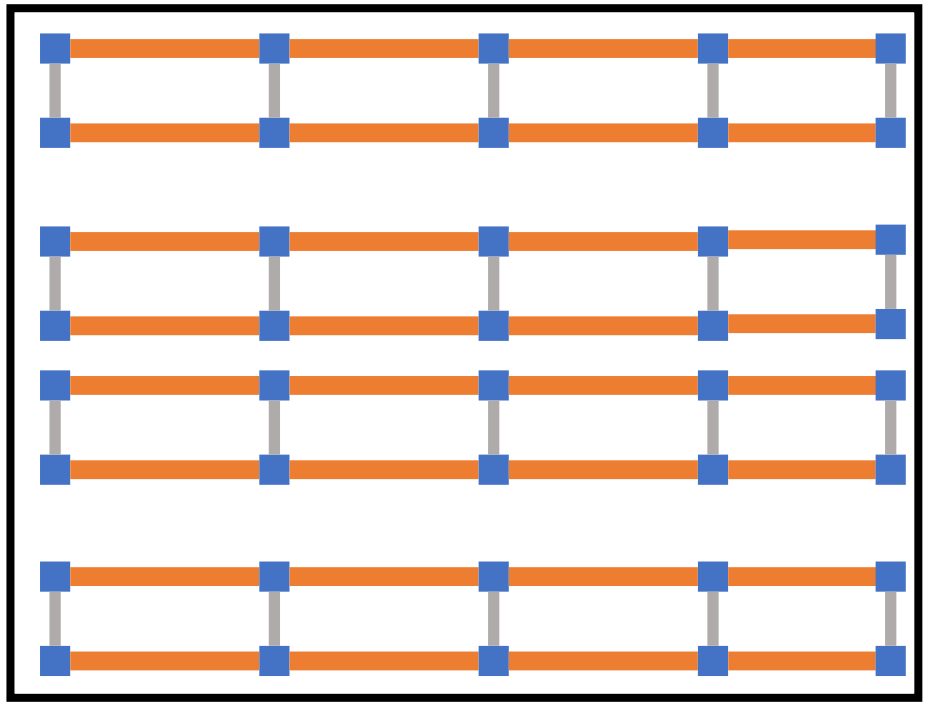
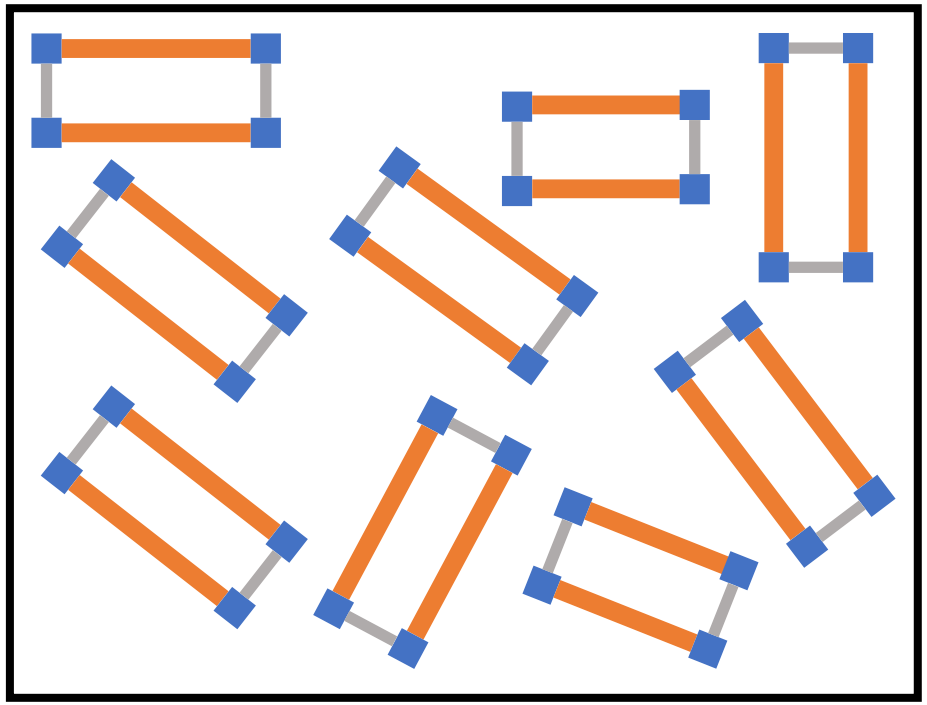
OUR PROBLEM



or



?

goal: **Place bays** in a warehouse the **cheapest way** using the largest ammount of

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INPUT

Each case consists on 4 CSV files

WAREHOUSE.CSV

OBSTACLES.CSV

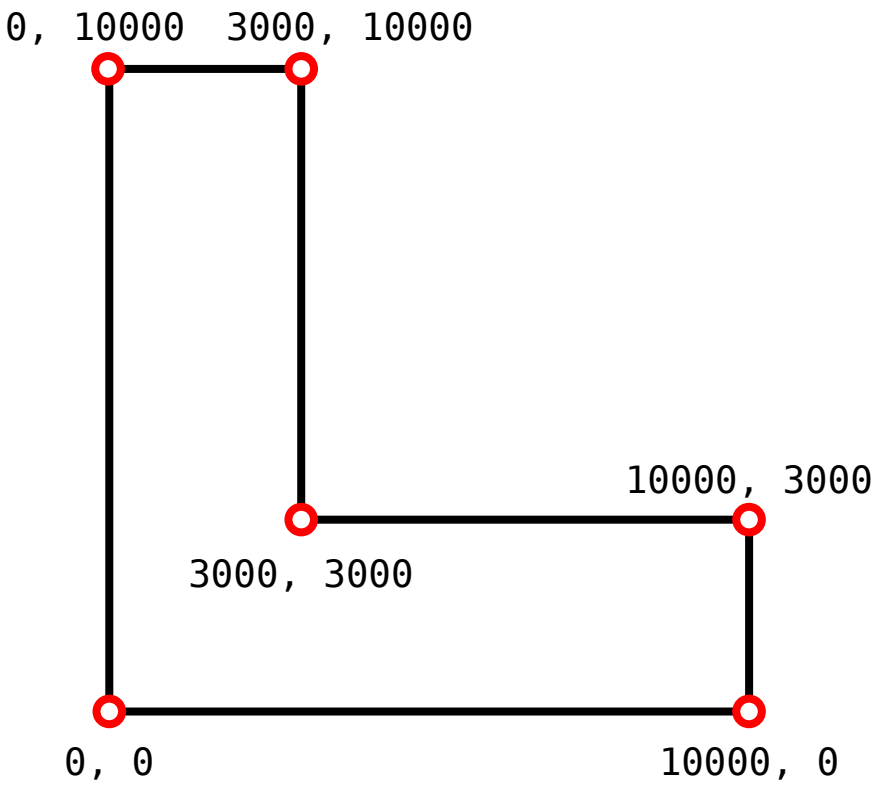
CEILING.CSV

TYPES_OF_BAYS.CSV

INPUT: WAREHOUSE

Coord X, Coord Y

```
0, 0
10000, 0
10000, 3000
3000, 3000
3000, 10000
0, 10000
```

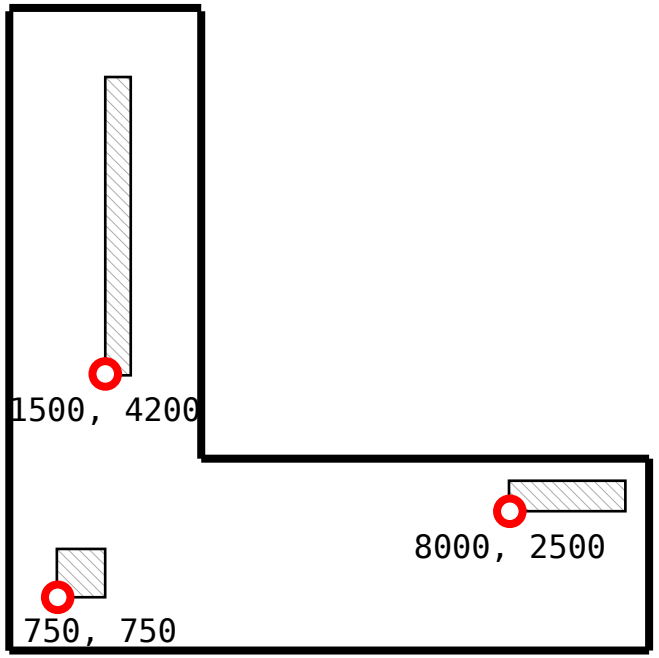


The walls of the warehouse will always be axis aligned

INPUT: OBSTACLES

Coord X, Coord Y, Width, Depth

```
750, 750, 750, 750
8000, 2500, 1500,
300
1500, 4200, 200,
4600
```



The obstacles will always be boxes

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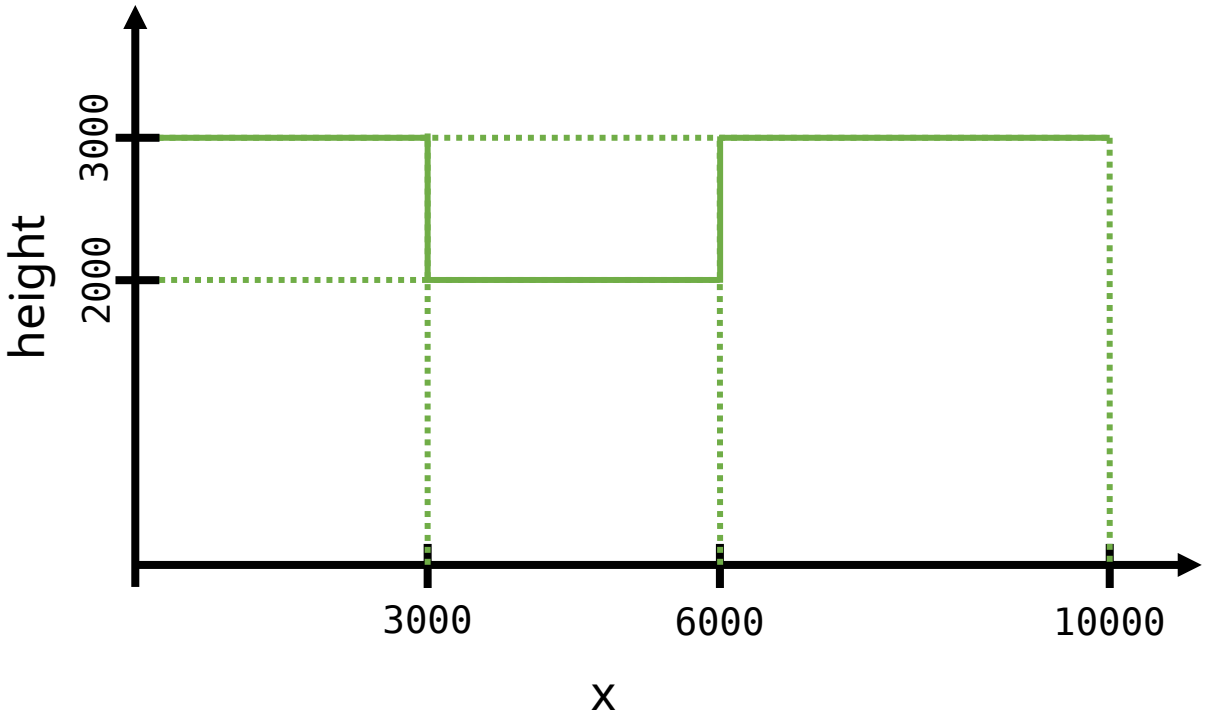
INPUT: CEILING

Coord X, Ceiling Height

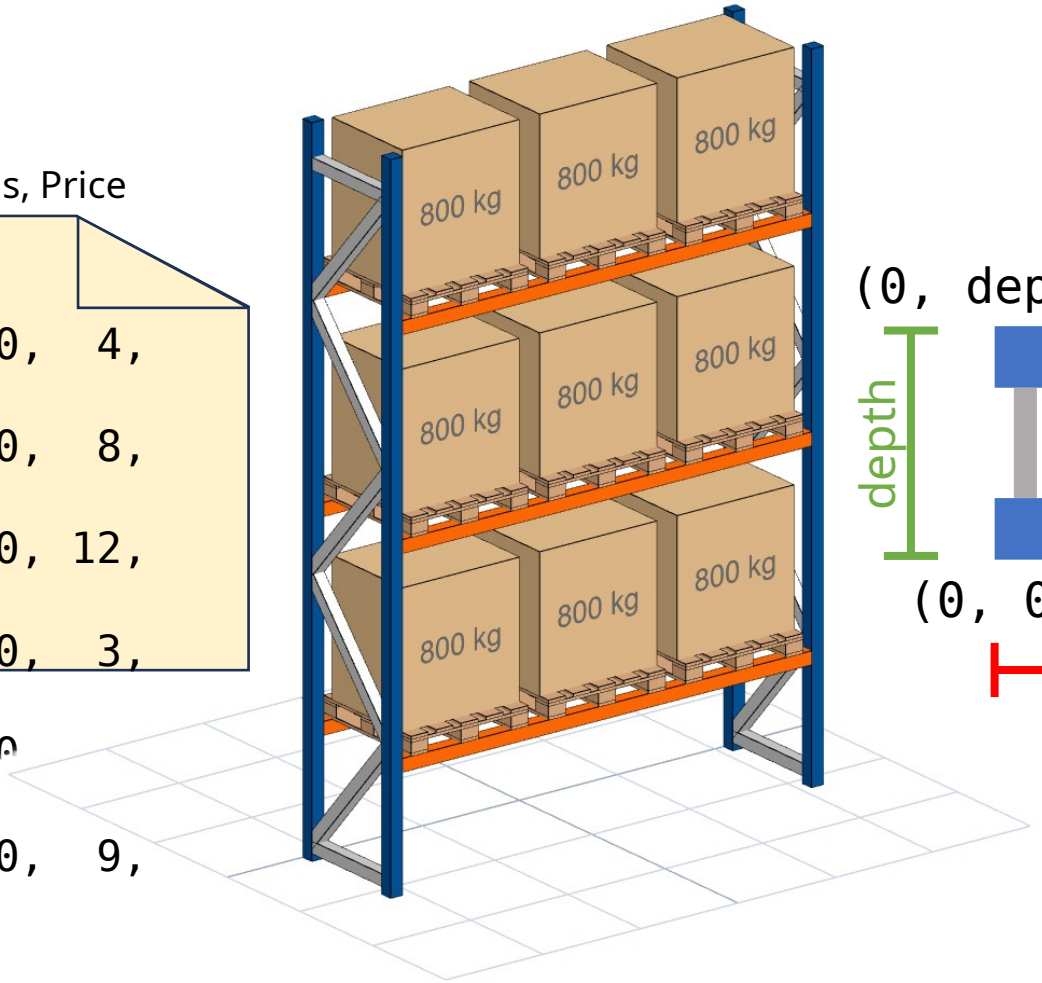
```
0, 3000
3000, 2000
6000, 3000
```

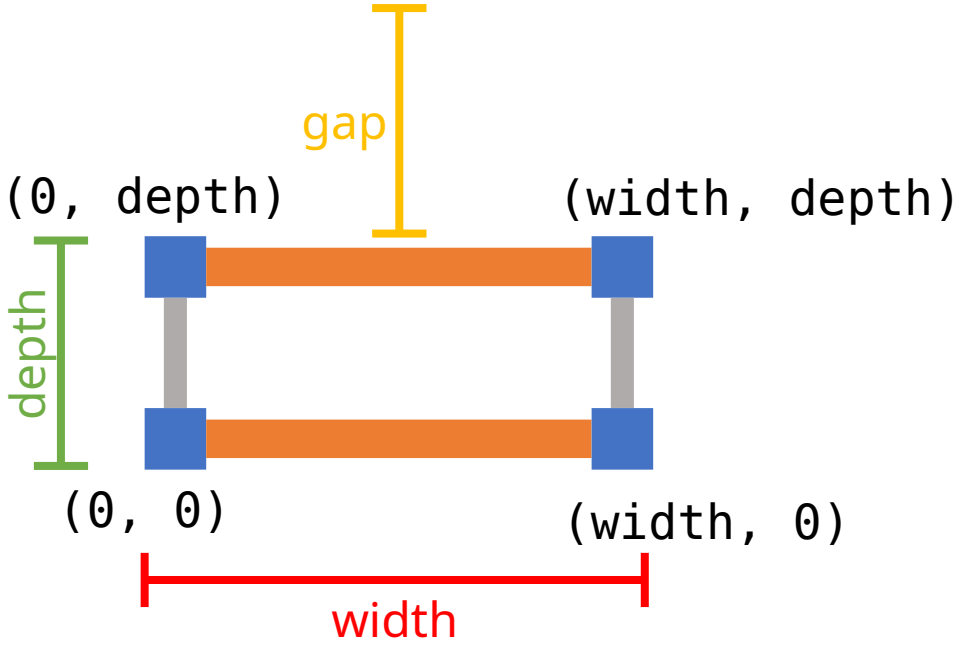


INPUT: BAY

Id, Width, Depth, Height, Gap, nLoads, Price

```
0,  800, 1200, 2800, 200,  4,
2000
1, 1600, 1200, 2800, 200,  8,
2500
2, 2400, 1200, 2800, 200, 12,
2800
3,  800, 1000, 1800, 150,  3,
1800
4, 1600, 1000, 1800, 150
2300
5, 2400, 1000, 1800, 150,  9,
2600
```





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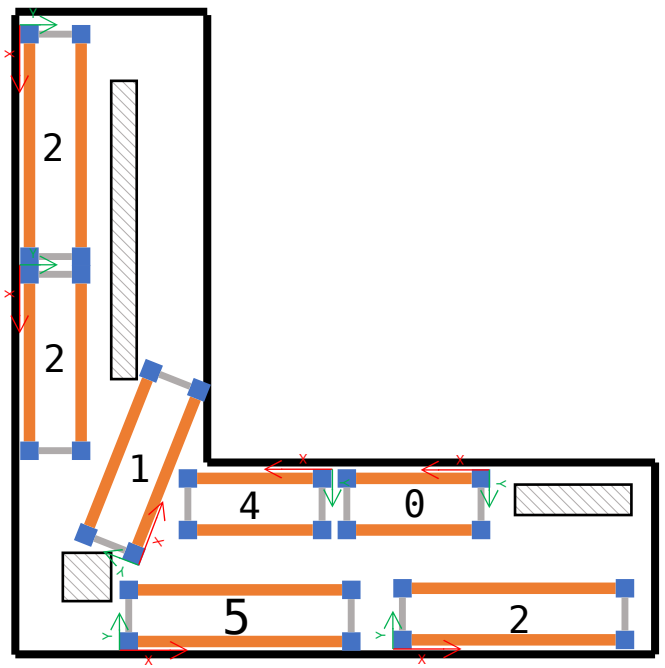
EXPECETED OUTPUT

Id, X, Y, Rotation

```
0, 6600, 2900, 180
1, 1000, 1600,  70
2,    0, 5400, 270
2,    0, 9900, 270
2, 5000,    0,   0
4, 4400, 2900, 180
5, 2000,    0,   0
```



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EVALUATION

Solution Generation

- Number of correct solutions
- Quality of the solution:

$$Q = \left(\sum_{bay} \frac{price}{loads} \right)^{2 - \frac{\sum_{bay} area}{area_{warehouse}}}$$

Presentation

- Visualization of the solution
- Whatever UI features

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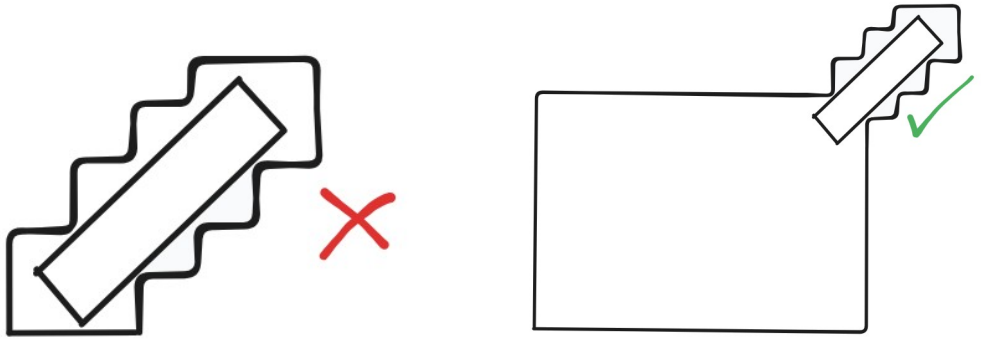
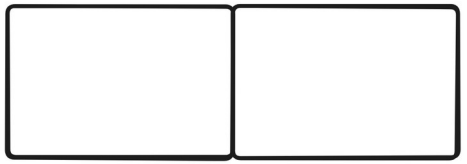
JUDGING

- During judging you will need to run some tests on your machine
  - Make sure it does not take more than 30s to finish

- Judging tests won’t be crazy edge cases.
  - They may be bigger than the examples



- Bays can share boundaries, same with warehouse
  - If a bay at position (0,0) has width 1000, you don’t need to put the next one at (1001,0), (1000,0) is fine.